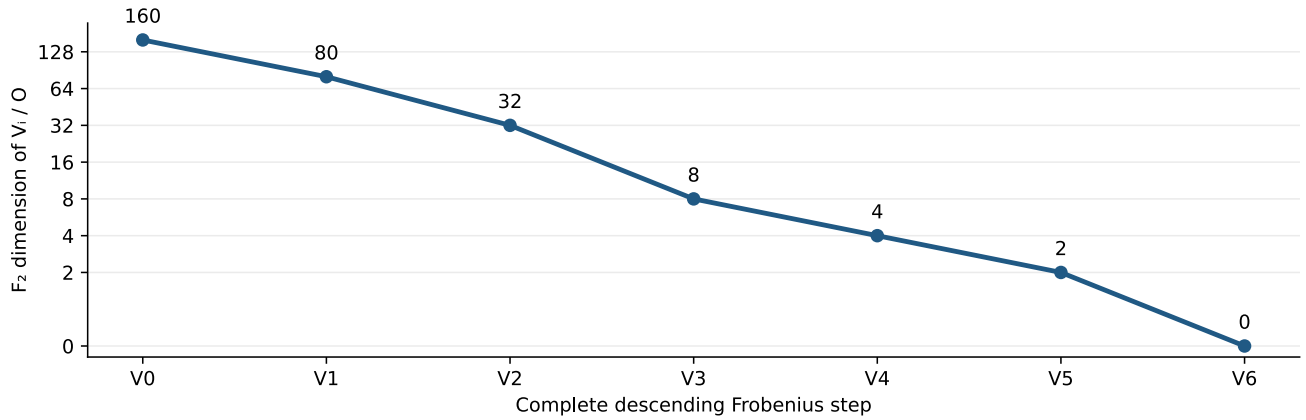
R70: affine order reaches the integral closure

The complete descending Frobenius chain on the rank-32 level-77 source order has binary quotient lengths $160, 80, 32, 8, 4, 2, 0$. The last two-class square map has 24 binary rows and rank two. Its first two rows are $(1, 0)$ and $(1, 1)$, a determinant-one witness that the kernel is zero. The sixth lattice is checked as the old order O by a full 32-by-32 field-coordinate identity and all 32 old-order basis squares.



Why this proves affine normality

The retained rank-32 order has all 528 source multiplication columns and an actual degree-32 field identification. Its nondegenerate trace Gram gives a trace-dual over-lattice V_0 containing every element integral over $F_2[z]$. The integral closure is square-closed, so it lies in each V_i by induction. As $V_6 = O$, it lies in O; conversely O lies in its integral closure. The corresponding conditional six-step implication is kernel-checked in Lean. The native source identities and trace-dual containment are the stated external premises.

Full attempt record and boundaries

The first native attempt is rejected despite a late marker: zero-column Singular errors occur after rank two. Fresh attempt 02 passes exit 0, the exact marker, zero error lines, and an explicit two-row witness. The first Lean compile failed on a field instance and set inclusion; the repaired second run compiled and audited its four authored claims. The third run added the direct six-step implication and audited 12 declarations with zero failing and zero trusted. All five attempts, including failures, are bound to historical source bytes in results.json.

This is the affine $F_2[z]$ chart. Proper branches, valuations of b, c, λ, ν and denominator functions at the omitted finite fibers and infinity, the level-77 map, and its order-11 action remain next. The general descending method is described by Singh–Swanson, Theorem 1.1.